

The Low-Power Atmospheric Compensation Experiment (LACE), also referred to as LOSAT-L and USA-51, was a military satellite developed by the Naval Research Laboratory for the United States' Strategic Defense Initiative in the late 1980s and early 1990s, otherwise referred to as the "Star Wars" program.

Background

The LACE mission concept began in February 1985 when the Strategic Defense Initiative Organization approached the Naval Research Laboratory to develop an experiment to characterize laser signals transmitted and received through the Earth's atmosphere from a ground station to an orbiting experiment. The laboratory had been involved with spaceflight since the dawn of the space race, having led the Navy's Vanguard program up until 1959. Since then the laboratory had developed satellites to perform experiments relating to solar radiation, radio transmission through the ionosphere, gravity gradient stabilization, and ocean surveillance, among others.

In July 1985, the Laser Communication Experiment (LACE) began development. The simple, spaceborne experiment was planned to be flown on the next flight of NASA's Long Duration Exposure Facility (LDEF), launched aboard the Space Shuttle. Because the LDEF was a completely passive payload, LACE would've required additional power and communication subsystems to function properly. After the Challenger disaster, shuttle launches were delayed indefinitely, and the LDEF would remain in orbit from the beginning of its first mission until 1990. This led to the further development of LACE into a full satellite by June 1986, and the availability of expendable launch vehicles such as the Atlas, Titan, and Delta rockets to launch government payloads such as LACE. The satellite would later be renamed to be the Low-power Atmospheric Compensation Experiment.

Before launch, LACE was placed in the Delta II's payload fairing alongside the Relay Mirror Experiment (RME) satellite. Also known as USA-52 and LOSAT-R, it was also sponsored by the

SDIO for ground-to-orbit laser tests and was built by Ball Aerospace. The two satellites and a third, LOSAT-X, were originally meant to be launched together, and were all part of the SDIO's LOSAT (LOW-altitude SATellite) program, even though they all had different designs and missions. LOSAT-X was removed from the launch manifest and would launch the following year with the GPS satellite USA-71.

Spacecraft

LACE did not have any onboard propulsion systems, instead, it relied on gravity-gradient stabilization to keep its experiments pointed towards Earth. It accomplished this using three 150 ft (46 m) long booms pointing along the forward, rear, and zenith axes of the spacecraft relative to the direction of its motion. While the zenith boom would remain fully extended during normal operation, the two booms along the forward and rear axes were designed to be extended and retracted 125 times during the 30 months of LACE's planned mission, and successfully did so over 65 times, albeit most movements were minor. The three booms were the largest retractable booms ever flown in space at the time of LACE's launch in 1990.

Instruments

Sensor Array Subsystem (SAS)

The Sensor Array Subsystem was the primary payload of the LACE satellite. It contained three sets of sensor arrays with a total of 210 sensors designed to detect laser emissions on-orbit from visible, pulsed, and infrared lasers on Earth.

The visible sensor array had 85 sensors distributed in the center of the earth-facing "target board". It was specifically designed to detect laser emissions from the Short Wavelength Adaptive Techniques (SWAT) program's argon ion laser at the Air Force Maui Optical Station (AMOS) in Maui, Hawaii. The array was sensitive to laser emissions between 400 nanometers (nm) to 1.06 micrometers (μm). Before launch it was calibrated for the SWAT laser's

wavelength of 514.5 nm, which was designed by the MIT Lincoln Laboratory. Later in the mission, this array was re-calibrated to 1.06 μm to support the USAF Starfire Optical Range at Kirtland AFB, New Mexico.

The pulsed array's 85 sensors were located in the same housings as the visible array sensors. They were calibrated at 354 nm and 1.06 μm , and were designed to generally support pulsed excimer lasers at wavelengths from 300 to 400 nm and a low-power laser emulator at 1.06 μm . The array detected pulses of durations from 10 nanoseconds (ns) to 2 microseconds (μs) with a maximum repetition rate of 100 pulses per second.

Like the visible array, the infrared array was also designed for a specific laser program: the Low Power Chemical Laser (LPCL) at White Sands, New Mexico. The array's 40 sensors were distributed uniformly across the target board, and detected deuterium fluoride chemical laser emissions between 3.6 and 4.0 μm .

Ultraviolet Plume Instrument (UVPI)

The purpose of the Ultraviolet Plume Instrument (UVPI) experiment was to accurately image and track the plume of a rocket launched from Earth's surface in the ultraviolet spectrum. It was a pointable telescope mounted on the earth-facing side of the LACE spacecraft. Its mission was to collect images of rocket plumes in the near-ultraviolet and midultraviolet regimes from a space-based platform. It was also used to collect background image data of Earth, Earth's limb, and celestial objects. Background imaging included the day, night, and dawn limb, aurora, sunlit and moonlit clouds, and the Earth's surface.

The UVPI consisted of two image-intensified CCD cameras that were boresighted and shared a Maksutov-Cassegrain telescope. The tracker camera was sensitive in the near-ultraviolet/visible wavelength region, had a field of view approximately 14 times larger in each dimension than the plume camera, and was used to locate, acquire, and track a target. The plume camera collected images at four wavelengths from 195 to 350 nm, selected with a

filter wheel. The field of view was 0.184 x 0.137 degrees. Its photometric range and sensitivity were optimized for nighttime operations. The tracker camera could image stars as dim as visual magnitude 7. Normal frame rate was 5 per second, but the zoom mode rate was 30 per second with a reduced field of view.

Army Background Experiment (ABE)

The Army Background Experiment was commissioned by the U.S. Army Strategic Defense Command to measure the space neutron background radiation from the LACE satellite. It consisted of four borated plastic scintillator rods 8 in. long by 3 in. in diameter attached to a fold-out panel on the nadir side of LACE. The rods would detect neutrons originating from atmospheric escape and from the spacecraft's body. The experiment was jointly developed by the Grumman Corporation and the Los Alamos National Laboratory, with the support of the U.S. Department of Energy. Within a few months after the launch of LACE, the ABE was reported to be operating nominally, and it was expected to remain in operation through LACE's expected 30 month lifetime.

Data from ABE was provided to the SDIO for the development of spaceborne systems to discriminate between warheads and decoys.

Radiation Detection Experiment (RDE)

The Radiation Detection Experiment (RDE) was an experiment on the earth-facing side of LACE to detect electromagnetic radiation. It was designed by The Aerospace Corporation for the U.S. Air Force. The classified experiment was the only one not operated for the Strategic Defense Initiative, and was not discussed by the SDIO nor the NRL prior to launch. It collected data for 14 months throughout LACE's mission.

Operational history

LACE was launched aboard a Delta II Rocket from Launch Complex 17B at Cape Canaveral Air Force Station on February 14, 1990, separating from the RME satellite approximately 15 minutes later. It was the first Department of Defense payload launched on a commercial launch vehicle.

About a month after launch, it was reported that LACE and RME had experienced some setbacks in their missions, but none critical enough to end their mission. A Pentagon official stated that LACE's retroreflector system wasn't sending back signals at the expected power level, which that official speculated that the issue could be due to heat damage to a reflector. LACE would proceed smoothly through the rest of its 60-day check-out phase, and it began its planned 30-month mission shortly after.

Exactly 3 years after its launch date, LACE was deactivated by the SDIO, ending its mission. It decayed into the atmosphere on May 24, 2000.

In 1998, while LACE was described as a "dead" satellite by the U.S. Army, its retroreflector was used as a target for the Low Power Chemical Laser and tracked with the SeaLite Beam Director at the High Energy Laser Systems Test Facility to support the DoD's MIRACL program.

The LACE program's net funding was \$122.3 million over its lifetime.

Plume observations

Space Shuttle

Several attempts were made between 1990 and 1992 to image the plumes fired by the Space Shuttle from its Orbital Maneuvering System pods and Reaction Control System thrusters as it orbited in low-earth orbit at a lower orbital inclination than LACE. The UVPI was listed as a secondary experiment in about half of the shuttle's missions launched during LACE's operation, including in pre-flight press kits and media resource kits, and post-flight mission summaries published by NASA. Shuttle astronauts were briefed on the UVPI experiment during training sessions before their missions, and they took photos of thruster firings on

board the shuttle done to support the UVPI, particularly on STS-50. However, no thruster firings were successfully imaged by the UVPI while the Space Shuttle was in orbit. There were a number of reasons why it was difficult for this to happen, according to "The NRL LACE Program Final Report":

Two bodies traveling at orbital velocities were involved.

The observation opportunity inevitably occurred away from LACE ground stations and had to be controlled by stored commands.

The shuttle could and did change its orbit after the controlling commands were stored in the LACE satellite.

NASA could never guarantee that there would be plume-generating activity during the few seconds of the observation opportunity.

For example, during the STS-44 mission, the SDIO planned to observe thruster firings from the Space Shuttle Atlantis's OMS and PRCS pods with LACE and the UVPI over four to six orbital conjunctions. However, due to Atlantis maneuvering to avoid a spent Soviet booster in orbit, and a failure of one of its IMUs, NASA's flight rules brought the mission to an early end.

Table of UVPI plume observation attempts

Gallery

Further reading

Fisher, Shalom; Schultz, Kenneth I.; Taylor, Lawrence W. (July 1995). "Vibrations of the Low Power Atmospheric Compensation Experiment satellite". *Journal of Guidance, Control, and Dynamics*. 18 (4): 650–656. Bibcode:1995JGCD...18..650F. doi:10.2514/3.21443. ISSN 0731-5090.

Walters, Wesley F. (June 1990). "Dynamic Analysis of the Low Power Atmospheric Compensation Experiment (LACE) Spacecraft" (PDF). Defense Technical Information Center. Monterey, CA: Naval Postgraduate School. Retrieved April 4, 2024.

Welch, Jeanne A.; Light, Bonnie; Trusty, Gary L.; Cosden, Thomas H. (October 24, 1991). "A Laser Test Set for the Low-Power Atmospheric Compensation Experiment Satellite" (PDF). Defense Technical Information Center. Washington, DC: Naval Research Laboratory. Retrieved June 3, 2024.

Notes

References

Citations

Space Shuttle Reports

External links

LACE at Gunter's Space Page

LACE at Encyclopedia Astronautica